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CSCI 6397 Software Testing

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Exam One Part II - Test Case Development

Specifications: Designing a vehicle toll discount system

Offers discounts on highway tolls based on three factors:

- **Vehicle Weight**: Measured in kilograms (weight).
- **Electric Vehicle (EV) Status**: Whether the vehicle is fully electric (isEV).
- **Carpool Status**: Whether the vehicle carries 3 or more passengers (isCarpool).
- **Invalid Input**: Any weight <= 0 must throw an IllegalArgumentException.
- **Tier 1 (0 < weight <= 1200)**:
  - 0% rebate regardless of EV or Carpool status (base rate for light compact vehicles).
- **Tier 2 (1200 < weight <= 3500)**:
  - 15% rebate if the vehicle is an EV AND Carpool.
  - 10% rebate if the vehicle is an EV OR Carpool.
  - 0% rebate if neither applies.
- **Tier 3 (weight > 3500)**:
  - 25% rebate if the vehicle is an EV AND Carpool.
  - Otherwise, 5% baseline "commercial green incentive" rebate.

➔ *Analyze Specifications → Identify Causes and Effects*

- List each distinct **cause** (input condition):

- ‘weight’
- ‘isEV’
- ‘isCarpool’

◆ List each distinct **effect** (output/action):

- return [double discountPercent]

➔ Identify Equivalence Partitions

◆ weight Partitions:

i. (-∞, 0]: Exception

ii. (0, 1,200]: Tier 1

iii. (1,200, 3,500]: Tier 2

iv. (3,500, ∞): Tier 3

➔ Identify Boundary Values

◆ weight BVA:

i. [INT_MIN]: Exception

ii. -1: Exception

iii. 0: Exception

iv. 1: Tier 1

v. 1,200: Tier 1

vi. 1,201: Tier 2

vii. 3,500: Tier 2

viii. 3,501: Tier 3

ix. [INT_MAX] Tier 3

➔ Build Decision Table

Cause	Rule #1	Rule #2	Rule #3	Rule #4	Rule #5	Rule #6
isEV	.	.	.	.	F	T
isCarpool	.	.	.	.	F	F
weight >= 0	F	T	T	T	T	T
weight >= 1	.	F	T	T	T	T
weight >= 1,200	.	F	F	T	T	T
weight >= 1,201	.	F	F	F	T	T
weight >= 3,500	.	F	F	F	.	.
weight >= 3,501	.	F	F	F	F	F
Effects	Exception UNUSED	0%	0%	0%	0%	10%

Cause	Rule #7	Rule #8	Rule #9	Rule #10	Rule #11
isEV	F	T	F	.	T
isCarpool	T	T	.	F	T
weight >= 0	T	T	T	T	T
weight >= 1	T	T	T	T	T
weight >= 1,200	T	T	T	T	T
weight >= 1,201	T	T	T	T	T
weight >= 3,500	.	.	T	T	T
weight >= 3,501	F	F	T	T	T
Effects	10%	15%	5%	5%	25%

➔ Build TCI Table

TCI	Partitions, Values, Rules
EP01	(-, 0]
EP02	(0, 1,200]
EP03	(1,200, 3,500]
EP04	(3,500, ∞)
BVA01	[INT_MIN]
BVA02	-1
BVA03	0
BVA04	1
BVA05	1,200
BVA06	1,201
BVA07	3,500
BVA08	3,501
BVA09	[INT_MAX]
DT01	Rule #2
DT02	Rule #3
DT03	Rule #4
DT04	Rule #5
DT05	Rule #6
DT06	Rule #7
DT07	Rule #8
DT08	Rule #9
DT09	Rule #10
DT10	Rule #11

➔ Build Test Cases

Test Case	TCI(s) Tested	weight	isEV	isCarpool	Expected Output
TC01	EP01, BVA01	[INT_MIN]	T	T	IllegalArgumentException
TC02	EP02, BVA04, DT02	1	F	F	0%
TC03	EP03, BVA07, DT04	3,500	F	F	0%
TC04	EP04, BVA08, DT08	3,501	F	T	5%
TC05	EP01, BVA02	-1	F	F	IllegalArgumentException
TC06	EP01, BVA03, DT01	0	F	F	IllegalArgumentException
TC07	EP02, BVA05, DT03	1,200	F	F	0%
TC08	EP03, BVA06, DT04	1,201	F	F	0%
TC09	EP04, BVA09, DT09	[INT_MAX]	T	F	5%
TC10	EP03, BVA06, DT05	1,201	T	F	10%
TC11	EP03, BVA06, DT06	1,201	F	T	10%
TC12	EP03, BVA06, DT07	1,201	T	T	15%

TC13	EP04, BVA08, DT10	3,501	T	T	25%